

SFWRENG 3IA3 - Practice Exam

Matthew Farah, farahm11@mcmaster.ca, 400468251

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Question 1

Let $X = (x_n)_{n=1}^{\infty}$ be a real valued sequence and let $L \in \mathbb{R}$.

(a) Define $\lim_{n \rightarrow \infty} x_n = L$.

Solution:

We say that $\lim_{n \rightarrow \infty} x_n = L$ if, for arbitrary $\epsilon > 0$, there exists some $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - L| < \epsilon$.

(b) Define X' is a subsequence of X .

Solution: